DAY 03 OF 10

Control Flow: Conditions & Loops

Make your programs decide and repeat

Today's agenda

1. if / elif / else
2. while loops
3. for loops and range()
4. break, continue, pass
5. Classic patterns

Python in 10 Days
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Making decisions with if

DAY 03

- if runs a block only when the condition is True
- elif checks the next condition; else is the fallback
- Do not forget the colon : and the indentation
- Only one branch runs

Python

```python
marks = 72
if marks >= 90:
    grade = "A"
elif marks >= 75:
    grade = "B"
elif marks >= 60:
    grade = "C"
else:
    grade = "F"
print("Grade:", grade)
```

OUTPUT

```
Grade: C
```

Nested if and one-line if

DAY 03

- You can put an if inside another if
- Ternary form: value_if_true if condition else value_if_false
- Truthy / falsy: 0, "", [], None are False; most other values are True
- Prefer clear code over clever one-liners

Python

```python
age = 20
has_id = True
if age >= 18:
    if has_id:
        print("Entry allowed")
    else:
        print("Show your ID")

status = "Adult" if age >= 18 else "Minor"
print(status)
```

OUTPUT

```
Entry allowed
Adult
```

The while loop

- Repeats while a condition stays True
- Make sure something inside changes the condition, or it never ends
- Great when you do not know the number of repeats in advance
- Stop an infinite loop with Ctrl + C

Python

```python
count = 1
while count <= 5:
    print(count, end=" ")
    count += 1
print()

# keep asking until correct
password = ""
while password != "python":
    password = input("Password: ")
```

OUTPUT

```
1 2 3 4 5
```

The for loop and range()

DAY 03

- for goes through each item of a sequence
- range(stop), range(start, stop), range(start, stop, step)
- The stop value is NOT included
- You can loop over strings, lists, dictionaries

Python

```python
for i in range(1, 6):
    print(i * i, end=" ")
print()
for n in range(10, 0, -3):
    print(n, end=" ")
print()
for ch in "Hi!":
    print(ch)
```

OUTPUT

```
1 4 9 16 25
10 7 4 1
H
i
!
```

break, continue and pass

- break: leave the loop immediately
- continue: skip the rest of this round and go to the next
- pass: do nothing (a placeholder for code you will write later)
- loop-else: else runs when the loop ends without break

Python

```python
for n in range(1, 11):
    if n % 2 == 0:
        continue      # skip even
    if n > 7:
        break         # stop
    print(n, end=" ")
```

OUTPUT

```
1 3 5 7
```

Pattern 1: sum and factorial

- Keep a running total in a variable
- Start sum at 0 and product at 1
- This 'accumulator' idea is used everywhere in data processing

Python

```python
total = 0
for n in range(1, 101):
    total += n
print(total)

fact = 1
for n in range(1, 6):
    fact *= n
print(fact)
```

OUTPUT

```
5050
120
```

Pattern 2: FizzBuzz

- A classic interview warm-up
- Multiples of 3 print Fizz, of 5 print Buzz, of both FizzBuzz
- Check the combined case first
- Shows if/elif ordering clearly

Python

```python
for n in range(1, 16):
    if n % 15 == 0:
        print("FizzBuzz")
    elif n % 3 == 0:
        print("Fizz")
    elif n % 5 == 0:
        print("Buzz")
    else:
        print(n)
```

Pattern 3: prime check and star pyramid

DAY 03

- Prime: only divisible by 1 and itself. Test divisors up to the square root
- Nested loops: the inner loop runs fully for each outer round
- Use end="" to stay on the same line

Python

```python
num = 29
is_prime = num > 1
for d in range(2, int(num ** 0.5) + 1):
    if num % d == 0:
        is_prime = False
        break
print(num, "prime" if is_prime else "not prime")

for i in range(1, 5):
    print(" " * (4 - i) + "*" * (2 * i - 1))
```

OUTPUT

```
29 prime
   *
  ***
 *****
*******
```

Practice time

DAY 03

1. Print the multiplication table of any number
2. Count the vowels in a sentence
3. Print all numbers from 1 to 100 divisible by 7
4. Number guessing game: pick 1-10, keep asking until correct
5. Reverse a number using a while loop (1234 becomes 4321)

Challenge: print the first 10 Fibonacci numbers

Day 3 recap

DAY 03

- if / elif / else make decisions
- while repeats on a condition; for walks through a sequence
- range(start, stop, step) generates numbers
- break leaves a loop, continue skips a round

Homework: finish practice. Next: Lists and Tuples